

5-Mark Questions & Answers

Q1. Explain the difference between File System and Database System with a diagram.

File System stores data in separate, unrelated files managed by the OS. Each application has its own set of files with no centralized control, leading to redundancy and inconsistency.

Database System centralizes all data under a DBMS, which enforces structure, integrity, and security across all applications.

Key Differences:

Feature	File System	Database System
Data Redundancy	Very High	Minimal (via normalization)
Data Consistency	Poor	High (single source of truth)
Security	OS-level only	Fine-grained (user/table/column level)
Concurrency	Not supported properly	Built-in concurrency control
Recovery	No automatic recovery	Transaction-based recovery (ACID)
Query Language	None (custom code required)	SQL
Data Sharing	Difficult	Easy and controlled

Diagram:

FILE S
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Pay
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HR
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DATABA
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Pay
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HR
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Q2. Explain the three-schema (three-level) architecture of DBMS with a neat diagram.

The Three-Schema Architecture (proposed by ANSI/SPARC) separates the database into three levels to achieve data abstraction and independence.

1. External Level (View Level):

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2. Conceptual Level (Logical Level):

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detail.

3. Internal Level (Physical Level):

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Diagram:

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Q3. Explain Data Abstraction and its three levels in detail with a diagram.

Data Abstraction is the process of hiding irrelevant implementation details from users to reduce complexity and improve usability.

1. Physical Level:

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2. Logical Level:

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3. View Level:

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Diagram:

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Q4. Explain Data Independence and its two types in detail with a diagram.

Data Independence is the ability to change the schema at one level without requiring changes at the next higher level. It is a key advantage of the three-level architecture.

Type 1 -- Logical Data Independence:

Ability

Type 2 -- Physical Data Independence:

Ability

Diagram:

VIEW L
|
LOGICA
|
PHYSIO

	Logical Independence	Physical Independence
Schema Changed	Conceptual	Internal/Physical
Level Protected	External views	Conceptual
Difficulty	Harder	Easier

Q5. Describe the different types of Database Users with examples.

1. Naive Users:

End-us

2. Application Programmers:

IT prof

3. Sophisticated Users:

Interac

4. Specialized Users:

Build s

5. Database Administrator (DBA):

Most p

Q6. Explain the role and responsibilities of a Database Administrator (DBA).

The DBA is the central authority of the database system with complete control and highest privilege level.

1. Schema Definition: Writes the original DDL to create the database schema -- tables, views, indexes, and integrity constraints.
2. Storage Structure and Access Method Definition: Defines physical data organization on disk. Chooses appropriate file organizations and indexing methods for optimal performance.
3. Schema and Physical Organization Modification: Alters the schema when business requirements change or modifies physical storage to improve performance.
4. Granting User Authorizations: Creates user accounts and uses GRANT/REVOKE to manage which users can perform which operations on which data.
5. Routine Maintenance: Performs periodic data backups, monitors disk space and query performance, applies software patches, and ensures high availability.
6. Performance Tuning: Analyzes slow queries, creates indexes, and restructures tables to ensure fast response times.
7. Defining Integrity Constraints: Adds rules like NOT NULL, UNIQUE, CHECK, FOREIGN KEY to maintain data accuracy and consistency.

Q7. Explain the components of the Storage Manager in detail.

The Storage Manager is the interface between the higher-level DBMS and physical data stored on disk.

Components:

1. Authorization and Integrity Manager: Checks user privileges before executing any query. Also enforces integrity constraints (PK, FK, NOT NULL).
2. Transaction Manager: Manages transactions -- logical units of work. Ensures the database remains consistent even after system failures. Enforces ACID properties. Manages the transaction log.

3. File Manager: Manages physical storage of data on disk. Allocates disk space, manages file organization (heap, sequential, indexed), and tracks data locations.

4. Buffer Manager: Manages main memory (RAM) buffer. Fetches disk pages into memory when needed and uses LRU (Least Recently Used) replacement policy to evict pages when memory is full, minimizing costly disk I/O.

Diagram:



Q8. Explain the Query Processor and its components in detail.

The Query Processor translates high-level user queries (SQL) into executable operations that the Storage Manager performs.

1. DDL Interpreter:

Interprets
Does not

2. DML Compiler:

Translates

3. Query Evaluation Engine:

Executes
disk.

Query Flow:



Q9. Explain the characteristics of a database system with examples.

1. Self-Describing Nature: A DBMS contains not just data but also a complete description (metadata) in a Data Dictionary, making it self-contained.
2. Data Abstraction: Hides physical storage details. Programmers work with logical tables, not raw files.
3. Controlled Redundancy: Minimizes data duplication through normalization, preventing inconsistency.
4. Multiple User Interfaces: Supports GUI forms for naive users, SQL for sophisticated users, APIs for programmers.
5. Data Sharing: Multiple users and applications can concurrently access the same database via concurrency control.
6. Data Security: Fine-grained privilege control (GRANT/REVOKE) prevents unauthorized access.
7. Backup and Recovery: Automatically logs transactions and restores the database to a consistent state after a crash.

Q10. Explain 1-tier, 2-tier, and 3-tier DBMS architecture with diagrams.

1-Tier Architecture:

All com

2-Tier Architecture:

Client

3-Tier Architecture:

Three

Q11. Differentiate between DBMS and File System (5 points) with a table.

Feature	File System	DBMS
Data Redundancy	Very high; same data copied in multipl	Minimal; data normalized and stored ce
Data Inconsistency	Common; updates in one file not refle	Eliminated; single update reflects sys
Data Access	No standard language; custom programs	Standard SQL supports flexible, ad-hoc
Security	Basic OS-level file permissions only	Fine-grained access control per user,
Concurrent Access	Unreliable; simultaneous writes can co	Built-in concurrency control with lock
Backup & Recovery	Manual; no automatic failure recovery	Automatic transaction logs and crash r
Data Integrity	Enforced manually within application c	Enforced automatically via DBMS constr
Cost	Cheap initially	Higher initial cost but lower long-ter

Example	` .txt`, `.csv` files managed by OS	MySQL, Oracle, PostgreSQL
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Q12. Explain Entity Sets and Relationship Sets in detail with examples and ER diagram notation.

1. Entity Sets

An Entity is an object in the real world that is distinguishable from other objects. An Entity Set is a set of entities of the same type that share the same properties.

2. Relationship Sets

A Relationship is an association among several entities. A Relationship Set is a set of relationships of the same type.

(Strong Entity)	\ /	(Strong Entity)
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Q13. Explain the different types of attributes (simple, composite, single-valued, multivalued, derived) with diagrams.

Composite		Multi-val		Attribute		Attribute
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1. Simple vs. Composite Attributes:

Q14. Explain Mapping Cardinality Constraints (1:1, 1:N, N:1, M:N) in detail with diagrams.

Mapping Cardinality defines the number of entities to which another entity can be associated via a relationship set:

- * One-to-One (1:1): Each entity in A connects to at most 1 entity in B, and vice versa. Example: EMPLOYEE *manages* DEPARTMENT.

Q15. Explain Weak and Strong Entity Sets in detail with a suitable example (e.g., Section-Department) and diagram.

Comparison Table

Feature	Strong Entity Set	Weak Entity Set
Primary Key	Has its own primary key	No primary key; has a Discriminator (P
Dependency	Independent existence	Existence-dependent on Owner Entity Se
Participation	Partial or Total	Always Total Participation in iden
ER Symbol	Single Rectangle	Double Rectangle
Relationship	Regular Relationship (Diamond)	Identifying Relationship (Double Diamo

Example: COURSE (Strong) and SECTION (Weak)

- * COURSE has primary key Course_ID.



Q16. Explain how Primary Keys are determined for entity sets and relationship sets, including all binary relationship cardinalities.

1. Primary Keys for Entity Sets

- * Strong Entity Set: Chosen from Candidate Keys (minimal superkey).

2. Primary Keys for Relationship Sets

Mapping Cardinality	Primary Key of Relationship Set	Explanation
Many-to-Many (M:N)	PK(A) + PK(B) (Union of both PKs)	Every pair is unique
One-to-Many (1:N)	PK(B) (Primary Key of Many side)	Each B maps to at most 1 A
Many-to-One (N:1)	PK(A) (Primary Key of Many side)	Each A maps to at most 1 B
One-to-One (1:1)	PK(A) OR PK(B) (Either one)	Unique on both sides

Q17. Explain Specialization and Generalization in the EER model with examples and diagrams.

- * Specialization (Top-Down): Breaking down a higher-level entity set into smaller, specialized lower-level entity sets based on distinguishing

traits.

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* Attribute Inheritance: Subclasses (STUDENT, EMPLOYEE) inherit all attributes (SSN, Name, Addr) from PERSON and add their own specific attributes.

Q18. Explain Total and Partial Participation constraints with examples and ER diagram notation.

* Total Participation (Existence Dependency): Every entity in the entity set MUST participate in at least one relationship instance. Drawn as a Double Line.

(Partial)	\ /	(Total)
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Q19. Explain Aggregation in the EER model with a suitable example (proj_guide, eval_for) and diagram.

Problem

In standard ER modeling, a relationship set cannot participate directly in another relationship set.

Solution

Aggregation abstracts a relationship set and its participating entity sets as a single higher-level entity set.

Example

Suppose EMPLOYEE works on PROJECT via relationship WORKS_ON. Now we need to record a MANAGER who evaluates this work via relationship EVALUATES.

	EMPLOYEE	-----< >-----	PROJECT	
+-----+ \ / +-----				
WORKS_ON				

+-----+-----+

+-----+

Q20. Explain Constraints on Generalization (Disjoint/Overlapping, Total/Partial) in detail.

Generalization/Specialization hierarchies are governed by two independent constraint dimensions:

1. Disjointness Constraint:

2. Completeness Constraint:

Q21. Draw and explain an ER diagram for a Teacher-Student "mentor" relationship, covering attributes, cardinality, and participation.

Structure

* Entity Sets: TEACHER(Teacher_ID, Name, Department), STUDENT(Roll_No, Name, Branch).

(Teacher_ID,Name)	1 \ / N	(Roll_No, Name)
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Q22. Explain the concept of Roles in relationships with the Course-Prerequisite example.

When an entity set participates more than once in the same relationship set (a recursive relationship), explicit Role Labels must be attached to the connecting lines in the ER diagram to prevent ambiguity.

Example: COURSE in PREREQUISITE

The PREREQUISITE relationship links COURSE back to COURSE.

(Course_ID, Title)

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Q23. Explain the steps for mapping a regular binary M:N relationship set to a relational schema, using WORKS_ON (Employee-Project).

Mapping Rule for M:N Relationship Set

1. Create a new separate relation (table) for the relationship set.

2. In

Example: EMPLOYEE (Emp_ID) M:N PROJECT (Proj_ID) via WORKS_ON (Hours)

- * EMPLOYEE table: EMPLOYEE(Emp_ID, Emp_Name, Salary) with PK = Emp_ID.

* PR

Q24. For the Banking System (Customer, Account, Transaction, Branch), draw the ER diagram and identify cardinalities.

Entities & Attributes

- * CUSTOMER(Cust_ID, Name, Address)

* AC

Relationships & Cardinalities

1. CUSTOMER *holds* ACCOUNT (M:N): A customer can hold multiple accounts (e.g., savings, current); an account can be jointly held by multiple customers.

2. BF

Q25. Explain the mapping of weak entity sets using the DEPENDENT-EMPLOYEE example. Why is the owner's primary key included as a foreign key

Mapping Steps

1. Create a relation for the owner entity set: EMPLOYEE(Emp_ID, Name, Salary) with PK = Emp_ID.

2. Cr

Why Owner's Primary Key is Included

A weak entity set lacks a primary key of its own (e.g., Dep_Name like "John" is not globally unique). Including Emp_ID grounds the weak entity to its specific owner, creating a composite primary key (Emp_ID, Dep_Name) that uniquely identifies each dependent record in the database.

Q26. Describe the mapping of a binary 1:N relationship set with an example, showing foreign key placement on the "N" side.

Mapping Rule for 1:N Relationship

Do NOT create a separate table. Instead:

Example: DEPARTMENT (1) *employs* EMPLOYEE (N)

* DEPARTMENT(Dept_ID, Dept_Name) with PK = Dept_ID.

Q27. Explain N-ary relationship mapping using the SUPPLIER-PROJECT-PART (SUPPLY) example. Why can't it be modeled as binary

N-ary Relationship Mapping Rule

Create a new relation for the N-ary relationship containing:

Example: SUPPLY (Degree 3 / Ternary)

* SUPPLIER(S_ID), PROJECT(P_ID), PART(Part_ID).

Why It Cannot Be Modeled as Binary

3 binary relationships (Supplier-Part, Supplier-Project, Part-Project) only state that Supplier S sells Part P, Supplier S works on Project PR, and Part P is used in Project PR. They cannot assert that *Supplier S supplied Part P specifically for Project PR in a single atomic transaction*.

Q28. For the College Admission System, design the ER diagram and convert it into a relational schema (relations with keys).

ER Components

* Entities: APPLICANT(App_ID, Name, Marks), COURSE(Course_ID, Title, Capacity).

Converted Relational Schema

1. APPLICANT (<u>App_ID</u>, Name, Marks)

Q29. Compare the four approaches to mapping specialization/generalization, stating when each applies.

Approach	Relational Schema Structure	Primary Key	Best Applied When
1. Superclass + Subclasses	1 table for superclass + 1 table per s	Superclass PK is PK for all tables	Generalization is overlapping/disjoint
2. Subclass-Only	Separate tables for each subclass only	Subclass PK	Generalization is **Total and Disjoint
3. Single Relation (1 Type Attr)	1 unified table with a Type discrimina	Superclass PK	Specialization is **Disjoint** with fe
4. Single Relation (Multi Type Attrs	1 unified table with boolean flags (is	Superclass PK	Specialization is **Overlapping with